



**BAULKHAM HILLS HIGH SCHOOL**

**2017**

**YEAR 12 HALF YEARLY  
HIGHER SCHOOL CERTIFICATE  
EXAMINATION**

# Mathematics Extension 2

## General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- All relevant mathematical reasoning and/or calculations must be shown

**Total marks – 70**

**Section I** (Pages 2-4)

**10 marks**

Attempt Questions 1-10

Allow about 15 minutes for this section

**Section II** (Pages 5-9)

**60 marks**

Attempt Questions 11-14

Allow about 1 hours 45 minutes for this section

## Section I

**10 marks**

**Attempt Questions 1-10**

**Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1 to 10.

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- 1 For the ellipse with the equation  $\frac{x^2}{16} + \frac{y^2}{25} = 1$ . What is the eccentricity of the ellipse?
- (A)  $\frac{9}{25}$   
(B)  $\frac{3}{5}$   
(C)  $\frac{1}{5}$   
(D)  $\frac{1}{\sqrt{5}}$
- 2 If  $w = \sqrt{3}(\cos \pi + i \sin \pi)$ , then  $w^4$  is equal to:
- (A)  $-9$   
(B)  $9$   
(C)  $81$   
(D)  $-81$
- 3 The polynomial  $P(z) = z^3 + (1 + i)z^2 + (1 + i)z + 1$  has a real zero  $z = -1$  and a complex zero  $z = \alpha$ . The third root is:
- (A)  $\frac{1}{\alpha}$   
(B)  $\bar{\alpha}$   
(C)  $-\alpha$   
(D)  $1 - \alpha$
- 4 What is the multiplicity of the root  $x = 1$  of the equation  $3x^5 - 5x^4 + 5x - 3 = 0$ ?
- (A) 1  
(B) 2  
(C) 3  
(D) 4

5  $z$  and  $w$  are two complex numbers. Which of the following statements is ALWAYS TRUE?

- (A)  $|z + w| \geq |z - w|$
- (B)  $|z + w| \leq |z - w|$
- (C)  $|z| - |w| \geq |z + w|$
- (D)  $|z| + |w| \geq |z - w|$

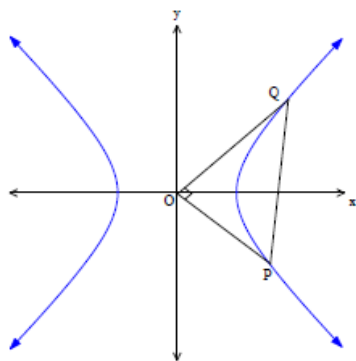
6 The polynomial  $P(x) = x^4 + ax^3 + bx^2 + cx + d$  has real coefficients and  $P(2i) = P(2 + i) = 0$ .

What is the value of  $a + b + c + d$ ?

- (A) 0
- (B) 6
- (C) 9
- (D) 49

7 The diagram below shows the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  where  $a > b > 0$ .

The points  $P(a \sec \theta, b \tan \theta)$  and  $Q(a \sec \alpha, b \tan \alpha)$  lie on the hyperbola and the chord  $PQ$  subtends a right angle at the origin.



Which of the following is true?

- (A)  $\tan \theta \tan \alpha = -\frac{a^2}{b^2}$
- (B)  $\tan \theta \tan \alpha = \frac{a^2}{b^2}$
- (C)  $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$
- (D)  $\sin \theta \sin \alpha = \frac{a^2}{b^2}$

8 If  $x^2 + 3xy = 5y^2$ , which of the following is an expression for  $\frac{dy}{dx}$ ?

(A)  $\frac{2x+3y}{10y-3x}$

(B)  $\frac{2x}{10y-9xy}$

(C)  $\frac{2x+3y}{\sqrt{5(x^2+3xy)}}$

(D)  $\frac{2x+3y}{10y}$

9 Seven travellers arrive in a town where there are 4 hotels. If two of the travellers want to stay in the same hotel, how many different accommodation arrangements are there?

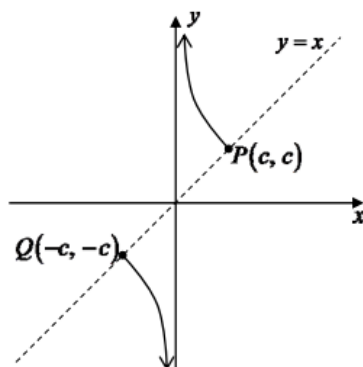
(A)  $\frac{6!}{2!2!} \times 2$

(B)  $\frac{7!}{2!2!2!} \times 2$

(C)  $4^6 \times 2$

(D)  $4^6$

10 The graph shows a part of the hyperbola  $xy = c^2$



Which pair of parametric equations precisely describes the sections of the hyperbola shown?

(A)  $x = c(t^2 + 1), y = \frac{c}{t^2+1}$

(B)  $x = c(1 - t^2), y = \frac{c}{1-t^2}$

(C)  $x = c\sqrt{1 - t^2}, y = \frac{c}{\sqrt{1-t^2}}$

(D)  $x = c \sin t, y = \frac{c}{\sin t}$

**End of Section I**

## Section II

**90 marks**

**Attempt questions 11 -14**

**Allow about 1 hours 45 minutes for this section**

Answer each question on the appropriate page of your answer booklet

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

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| Question 11 (15 marks) |                                                                                                                                | Marks |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------|
| (a)                    | Let $z = 2 + i$ and $w = i - 1$ . Find, in the form $x + iy$ :                                                                 |       |
|                        | (i) $3z + iw$                                                                                                                  | 1     |
|                        | (ii) $z\bar{w}$                                                                                                                | 2     |
| (b)                    | (i) Express $\sqrt{3} + i$ in modulus argument form.                                                                           | 1     |
|                        | (ii) If $z = \sqrt{3} + i$ , hence show that $z^7 + 64z = 0$ .                                                                 | 2     |
| (c)                    | Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$ .                                                |       |
|                        | (i) Find the eccentricity of the hyperbola.                                                                                    | 1     |
|                        | (ii) Find the coordinates of the foci and the $x$ -intercepts.                                                                 | 2     |
|                        | (iii) Find the equations of the directrices and the equations of the asymptotes.                                               | 2     |
| (d)                    | Sketch the region in the complex plane where the inequalities $ z + \bar{z}  \leq 1$ and $ z - i  \leq 1$ hold simultaneously. | 2     |
| (e)                    | Let $\omega$ be one of the complex roots of the equation $z^3 - 64 = 0$ .                                                      |       |
|                        | (i) Show that $\omega^2 = -4(\omega + 4)$ .                                                                                    | 1     |
|                        | (ii) Hence evaluate $(\omega + 4)^3$ .                                                                                         | 1     |
| End of Question 11     |                                                                                                                                |       |

**Question 12 (15 marks)****Marks**

(a) The roots of  $x^3 + 6x^2 + 5x - 2 = 0$  are  $\alpha, \beta$  and  $\gamma$ .

(i) Find the monic polynomial with roots  $\alpha^2, \beta^2$  and  $\gamma^2$ .

**2**

(ii) Find the value of  $\alpha^3 + \beta^3 + \gamma^3$ .

**3**

(b) Find a possible polynomial equation  $P(x)$ , of smallest degree that satisfies the following conditions:

**3**

- The polynomial  $P(x)$  has rational coefficients.
- Two of its roots are  $1 + \sqrt{5}$  and  $-6i$ .
- When  $y = P(x)$  is drawn on a Cartesian plane, there are four  $x$ -intercepts.

(c) A complex number  $\omega$  is such that  $|\omega| = 1$ .

**2**

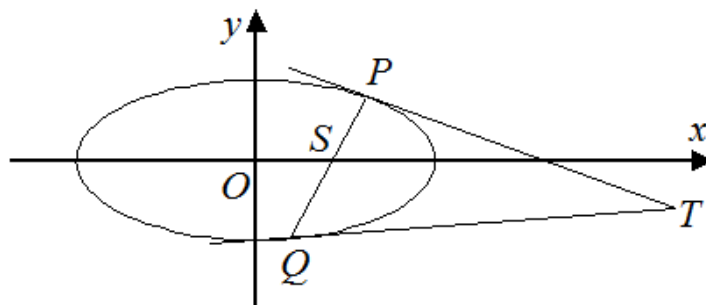
If  $z = \frac{1+\omega}{1-\omega}$ , find the locus of  $z$  as  $\omega$  moves on the complex number plane.

(d) The point  $P$  lies on the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , where  $a > b$

The chord through  $P$  and the focus  $S(ae, 0)$  meets the ellipse again at  $Q$ .

The tangents to the ellipse at  $P$  and  $Q$  meet at the point  $T(x_0, y_0)$ .

The equation of  $PQ$  is  $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ . **[DO NOT PROVE THIS]**



(i) Show that  $T$  lies on the directrix.

**1**

The point  $P$  is now chosen so that  $T$  lies on the  $x$ -axis.

(ii) What is the value of the ratio  $PS:ST$ ?

**1**

(iii) Show that  $\angle PTQ$  is acute.

**1**

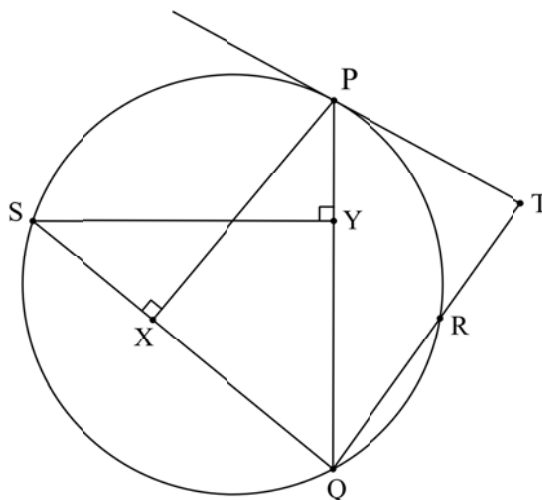
(iv) Show that the area of the triangle  $PQT$  is  $b^2 \left( \frac{1}{e} - e \right)$ .

**2****End of Question 12**

**Question 13 (15 marks)**

**Marks**

- (a) In the diagram below,  $TP$  is the tangent of the circle at  $P$ , and  $TQ$  is a secant cutting the circle at  $R$ .  
 $SQ$  is a chord of the circle such  $PX$  and  $SY$  are perpendicular to  $SQ$  and  $PQ$  respectively.

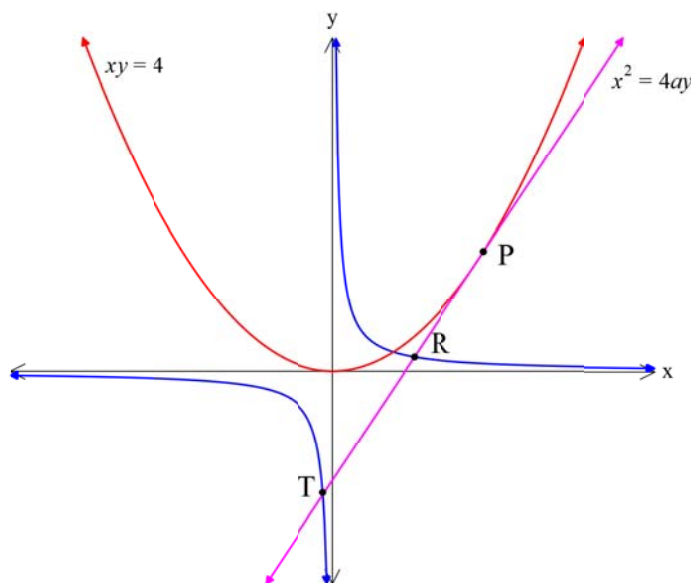


**Copy the diagram in your answer booklet.**

- |                                                                                                 |   |
|-------------------------------------------------------------------------------------------------|---|
| (i) Prove that $\angle TRP = \angle TPQ$ .                                                      | 2 |
| (ii) Explain why $SPYX$ is a cyclic quadrilateral and state the diameter of the circle $SPYX$ . | 2 |
| (iii) Prove $\angle PYX = \angle PRQ$ .                                                         | 2 |

**Question 13 continues on the next page**

- (b)  $P(8p, 4p^2)$  is a point on the parabola  $x^2 = 16y$ .  
The tangent to the parabola at  $P$  cuts the hyperbola  $xy = 4$  at two distinct points  $R$  and  $T$ .



- (i)  $M$  is the midpoint of  $RT$ , show that  $M$  has coordinates  $(2p, -2p^2)$ . 2  
You may assume the tangent to the parabola at  $P$  is  
 $y = px - 4p^2$ . **[DO NOT PROVE THIS]**
- (ii) Find the equation of the locus of  $M$ , as  $P$  moves on the parabola  $x^2 = 16y$ , stating any restrictions. 3

- (c) Let  $z = x + iy$  be any non-zero complex number.

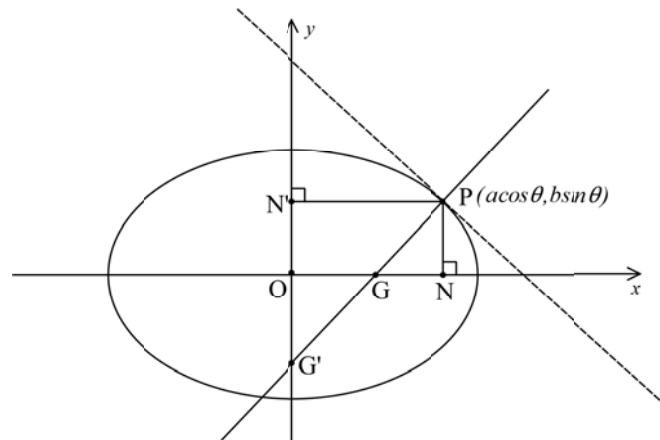
- (i) Show  $z + \frac{1}{z} = x + \frac{x}{x^2+y^2} + i\left(y - \frac{y}{x^2+y^2}\right)$ . 1
- (ii) Given that  $z + \frac{1}{z} = k$ , where  $k$  is real, show that  $y = 0$  or  $x^2 + y^2 = 1$ . 1
- (iii) Show also that if  $y = 0$  then  $|k| \geq 2$  and that if  $x^2 + y^2 = 1$  then  $|k| \leq 2$ . 2

**End of Question 13**

**Question 14 (15 marks)**

**Marks**

- (a) The diagram below shows the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , where  $a > b$ .  
The point  $P(a \cos \theta, b \sin \theta)$  lies on the ellipse.



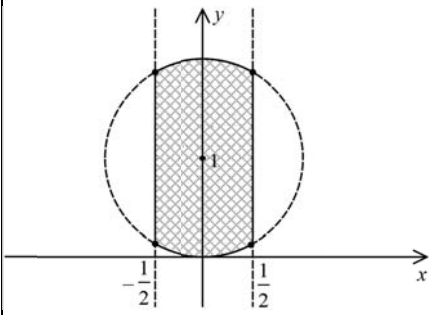
The normal to the ellipse at  $P$  meets the major and minor axes of the ellipse at  $G$  and  $G'$  respectively.  $N$  and  $N'$  are the feet of the perpendiculars from  $P$  to the major axes and minor axes respectively.

- (i) Show that the equation of the normal at  $P$  is  $ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2$ . 2
- (ii) Show that the ratio of  $OG : ON = e^2 : 1$ . 2
- (iii) Given that  $\Delta PN'G' \sim \Delta PNG$ , find the ratio of the area of  $\Delta PN'G' : \Delta PNG$ . 1
- (b) (i) Prove  $4 \cos \frac{\theta}{2} \cos \frac{\theta}{4} \sin \frac{\theta}{4} = \sin \theta$  1
- (ii) Prove by Mathematical Induction that for any positive integer  $n$  3
- $$\sin \theta = 2^n \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \sin \frac{\theta}{2^n}$$
- (iii) Explain why 1
- $$\lim_{n \rightarrow \infty} \left( \frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \times 2^n \right) = 1$$
- (iv) Hence using (ii), (iii) and choosing a suitable value for  $\theta$ , show: 3
- $$\frac{2}{\pi} = \cos \frac{\pi}{4} \times \cos \frac{\pi}{8} \times \cos \frac{\pi}{16} \times \dots$$
- (v) Show that  $\pi = 2 \left( \frac{2}{\sqrt{2}} \times \frac{2}{\sqrt{2+\sqrt{2}}} \times \dots \right)$  2

**End of Exam**

**HSC Year12 Extension 2 Half Yearly 2017**

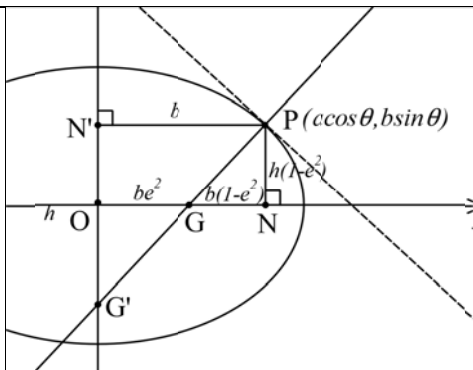
|   | Solutions                                                                                                                                                                                                                                                                                                                                                                                     | Mks | Comments |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 1 | $b^2 = a^2(1 - e^2)$ $16 = 25(e^2 - 1)$ $e = \frac{3}{5}$                                                                                                                                                                                                                                                                                                                                     | 1   | B        |
| 2 | $w^4 = \sqrt[4]{3} \operatorname{cis}(4 \times \pi)$ $w^4 = 9 \operatorname{cis} 4\pi$ $w^4 = 9 \operatorname{cis} 0$ $w^4 = 9$                                                                                                                                                                                                                                                               | 1   | B        |
| 3 | <p>Let the roots be <math>\alpha, \beta, -1</math></p> <p>Product of roots = <math>-\frac{b}{a}</math></p> $\alpha \times \beta \times -1 = -1$ $\beta = \frac{1}{\alpha}$                                                                                                                                                                                                                    | 1   | A        |
| 4 | <p>Let <math>P(x) = 3x^5 - 5x^4 + 5x - 3</math></p> $P(1) = 0$ $P'(x) = 15x^4 - 20x^3 + 5$ $P'(1) = 0$ $P''(x) = 60x^3 - 60x^2$ $P''(1) = 5$ $P'''(x) = 180x^2 - 120x$ $P'''(1) = 60$ <p><math>\therefore P(x)</math> has multiplicity 3</p>                                                                                                                                                  | 1   | C        |
| 5 | Triangular inequality                                                                                                                                                                                                                                                                                                                                                                         | 1   | D        |
| 6 | <p>The roots are <math>2i, -2i, 2 + i, 2 - i</math></p> $P(x) = (x^2 + 4)(x^2 - 4x + 5)$ $P(x) = x^4 - 4x^3 + 9x^2 - 16x + 20$ $a + b + c + d = -4 + 9 - 16 + 20$ $= 9$                                                                                                                                                                                                                       | 1   | C        |
| 7 | $m_{OP} = \frac{b \tan \theta}{a \sec \theta}$ $m_{OQ} = \frac{b \tan \alpha}{a \sec \alpha}$ $m_{OP} \times m_{OQ} = -1$ $\frac{b \tan \theta}{a \sec \theta} \times \frac{b \tan \alpha}{a \sec \alpha} = -1$ $\frac{\cos^2 \theta}{\cos^2 \theta} \times \frac{b^2 \tan \theta}{a^2 \sec \theta} \times \frac{\tan \alpha}{\sec \alpha} = -1$ $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$ | 1   | C        |
| 8 | <p>Derive both side implicitly:</p> $2x + \left(3 \times y + \frac{dy}{dx} \times 3x\right) = 10y \times \frac{dy}{dx}$ $2x + 3y = \frac{dy}{dx}(10y - 3x)$ $\frac{dy}{dx} = \frac{2x + 3y}{10y - 3x}$                                                                                                                                                                                        | 1   | A        |
| 9 | <p>If two particular people are together, then there are 6 entities, ie (AB)CDEFG.</p> <p>(AB) will have 4 rooms to choose from</p> <p>C will have 4 rooms to choose from</p> <p>...</p> <p>G will have 4 rooms to choose from</p> <p><math>\therefore</math> there are <math>4^6</math> ways to place 6 entities.</p>                                                                        | 1   | D        |

|          | Solutions                                                                                                                                                                                                                                                                                                                          | Mks | Comments                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10       | (A) False as $x$ is positive, and there are $x$ -ordinates that are negative<br>(B) False as $t \rightarrow \infty, x \rightarrow -\infty$<br>(C) False as $x$ is positive, and there are $x$ -ordinates that are negative<br>(D) True as $-1 \leq \sin t \leq 1, -c \leq c \sin t \leq c$ ,<br>ie $-c \leq x$ - ordinate $\leq c$ | 1   | D                                                                                                                                                                                                                               |
| 11a(i)   | $3(2+i) + i(i-1) = 6 + 3i - 1 - i$ $= 5 + 2i$                                                                                                                                                                                                                                                                                      | 1   | <b>1 mark</b><br>• correct answer.                                                                                                                                                                                              |
| 11a(ii)  | $(2+i)(-1-i) = -1 - 3i$                                                                                                                                                                                                                                                                                                            | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• correct conjugate<br>• correct multiplication with a wrong expression of the conjugate.                                                                              |
| 11b(i)   | $z = 2 \left[ \cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right) \right]$                                                                                                                                                                                                                                           | 1   | <b>1 mark</b><br>• correct answer.                                                                                                                                                                                              |
| 11b(ii)  | $z^7 - 64z = z(z^6 - 64)$ $z^6 = 2^6 \left( \cos\frac{6\pi}{6} + i \sin\frac{6\pi}{6} \right)$ $z^6 = 64$ $z^7 - 64z = z(64 - 64)$ $= 0$                                                                                                                                                                                           | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• correct use of DM's Thm                                                                                                                                              |
| 11c(i)   | $9 = -16(1 - e^2)$ $e = \frac{5}{4}$                                                                                                                                                                                                                                                                                               | 1   | <b>1 mark</b><br>• correct answer                                                                                                                                                                                               |
| 11c(ii)  | $\therefore \text{foci are } (\pm 5, 0)$ $x - \text{int when } y = 0$ $\frac{x^2}{16} = 1$ $x\text{-intercepts are } (\pm 4, 0)$                                                                                                                                                                                                   | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• correct foci<br>• correct intercept                                                                                                                                  |
| 11c(iii) | Directrices:<br>$x = \pm \frac{16}{5}$ Eqn of asymptote:<br>$y = \pm \frac{3}{4}x$                                                                                                                                                                                                                                                 | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• correct directrices<br>• correct asymptote                                                                                                                           |
| 11d      | $ z + \bar{z}  \leq 1$ $ 2x  \leq 1$ $-\frac{1}{2} \leq x \leq \frac{1}{2}$                                                                                                                                                                     | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• Shows region inside circle at centre (0,1)<br>• Shows region between the lines $-\frac{1}{2} \leq x \leq \frac{1}{2}$<br>• Correct boundaries with incorrect region. |
| 11e(i)   | $\omega^3 - 64 = 0$ $(\omega - 4)(\omega^2 + 4\omega + 16) = 0$ Since $\omega$ is a complex root so $\omega \neq 4$ $\omega^2 + 4\omega + 16 = 0$ $\omega^2 = -4\omega - 16$ $\omega^2 = -4(\omega + 4)$                                                                                                                           | 1   | <b>1 mark</b><br>• correct solution.                                                                                                                                                                                            |

|         | Solutions                                                                                                                                                                                                                                                                                                                                        | Mks | Comments                                                                                                                                                                                                                                                                             |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11e(ii) | $(\omega + 4)^3 = \left(\frac{\omega^2}{-4}\right)^3$ $= \frac{(\omega^3)^2}{-64}$ $= \frac{(64)^2}{-64}$ $= -64$                                                                                                                                                                                                                                | 1   | <b>1 mark</b><br>• correct solution.                                                                                                                                                                                                                                                 |
| 12a(i)  | Let $A = x^2$ , where $x = \alpha, \beta$ and $\gamma$<br>$x = \sqrt{A}$ $\sqrt{A}^3 + 6\sqrt{A}^2 + 5\sqrt{A} - 2 = 0$ $A\sqrt{A} + 5\sqrt{A} = 2 - 6A$ Square both sides<br>$A(A + 5)^2 = (2 - 6A)^2$ $A^3 - 26A^2 + 49A - 4 = 0$ $\therefore x^3 - 26x^2 + 49x - 4 = 0$                                                                       | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• substitutes $x\sqrt{A}$                                                                                                                                                                                                   |
| 12a(ii) | $\alpha^3 + 6\alpha^2 + 5\alpha - 2 = 0$ $\beta^3 + 6\beta^2 + 5\beta - 2 = 0$ $\gamma^3 + 6\gamma^2 + 5\gamma - 2 = 0$ $\alpha^3 + \beta^3 + \gamma^3 + 6(\alpha^2 + \beta^2 + \gamma^2) + 5(\alpha + \beta + \gamma) - 2 \times 3 = 0$ $\alpha^3 + \beta^3 + \gamma^3 + 6(26) + 5(-6) - 2 \times 3 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = -120$ | 3   | <b>3 mark</b><br>• correct solution.<br><b>2 mark</b><br>• significant progress to finding $\alpha^3 + \beta^3 + \gamma^3$<br><b>1 mark</b><br>• Finds $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha + \beta + \gamma$<br>• Forms a polynomial with $\alpha^2, \beta^2$ and $\gamma^2$ |
| 12b     | If the polynomial has rational coefficients, then the roots must be $-6i, 6i, 1 + \sqrt{5}, 1 - \sqrt{5}$<br>As the equation $(x^2 + 36)(x^2 - 2x - 4)$ has two x-intercepts, it needs any two factors to have 4 x intercepts. The smallest degree must be 6.<br>A possible equation is $x(x - 1)(x^2 + 36)(x^2 - 2x - 4)$                       | 3   | <b>3 mark</b><br>• correct solution.<br><b>2 mark</b><br>• Two conditions met.<br><b>1 mark</b><br>• One condition met                                                                                                                                                               |
| 12c     | $z - z\omega = 1 + \omega$ $z - 1 = \omega(1 + z)$ $ z - 1  =  \omega  \times  1 + z $ But $ \omega  = 1$<br>$ z - 1  =  z + 1 $ $\therefore \text{locus is } x = 0$                                                                                                                                                                             | 2   | <b>2 mark</b><br>• correct solution.<br><b>1 mark</b><br>• eliminates $\omega$<br>• simplifies $z$ using $ \omega  = 1$                                                                                                                                                              |
| 12d (i) | $PQ$ passes through $S(ae, 0)$<br>$\frac{ae x_0}{a^2} + \frac{0 \times y_0}{b^2} = 1$ $\therefore x_0 = \frac{a}{e}$ So $T(x_0, y_0)$ lies on the directrix $x = \frac{a}{e}$                                                                                                                                                                    | 1   | <b>1 mark</b><br>• correct solution                                                                                                                                                                                                                                                  |



|          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mks                                                                                                                                                                                                  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13b(iii) | <p>Sub <math>p = \frac{x_M}{2}</math> into <math>y_M</math></p> $y_M = -2 \times \left(\frac{x_M}{2}\right)^2$ <p><math>\therefore</math> The locus of <math>M</math> lies on <math>y = -\frac{x^2}{2}</math></p> <p>At (0,0) the tangent at <math>P</math> does not touch the hyperbola.</p> <p><math>\therefore x \neq 0</math></p> <p>Since <math>P</math> and <math>Q</math> is distinct, <math>P \neq Q</math></p> $(2) \rightarrow y = \frac{4}{x}$ <p>sub (2) <math>\rightarrow</math> (1)</p> $\frac{4}{x} = px - 4p^2$ $px^2 - 4p^2x - 4 = 0$ $\Delta > 0$ $16p^4 - 4 \times p \times -4 > 0$ $16p(p^3 + 1) > 0$ $16p \geq 0 \text{ or } p^3 < -1$ $4x > 0 \text{ or } \frac{x}{2} < -1$ $x > 0 \text{ or } x < -2$ <p><u><math>\therefore x &lt; -2 \text{ or } x &gt; 0</math></u></p> | 3                                                                                                                                                                                                    | <p><b>3 marks</b></p> <ul style="list-style-type: none"><li>• correct solution.</li></ul> <p><b>2 mark</b></p> <ul style="list-style-type: none"><li>• finds any part of the correct restriction</li><li>• correct inequality and solves.</li></ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• finds the equation of M lies on <math>y = -\frac{x^2}{2}</math></li><li>• finds <math>16p(p^3 + 1) &gt; 0</math></li></ul> |                                                                                                                                                                                                         |
| 13c(i)   | $z + \frac{1}{z} = x + iy + \frac{1}{x + iy}$ $= x + iy + \frac{x - iy}{x^2 + y^2}$ $= x + \frac{x}{x^2 + y^2} + i\left(y - \frac{y}{x^2 + y^2}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                    | <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• correct solution</li></ul>                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                         |
| 13c(ii)  | <p>If <math>k</math> is real then</p> $\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$ $y - \frac{y}{x^2 + y^2} = 0$ $y(x^2 + y^2) - y = 0$ $y(x^2 + y^2 - 1) = 0$ $y = 0 \text{ or } x^2 + y^2 = 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                    | <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• correct solution</li></ul>                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                         |
| 13c(iii) | <p>If <math>y = 0</math></p> $x + \frac{x}{x^2 + 0^2} = k$ $x + \frac{1}{x} = k$ $x^2 + 1 = kx$ $x^2 - kx + 1 = 0$ $\Delta \geq 0$ $k^2 - 4 \geq 0$ $k^2 \geq 4$ $ k  \geq 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>If <math>x^2 + y^2 = 1</math></p> $x + x = k$ $x = \frac{k}{2}$ <p>Since <math>x^2 + y^2 = 1</math>,</p> <p>The <math>x</math>-ordinate must be <math>\leq 1</math></p> $ x  \leq 1$ $ k  \leq 2$ | 2                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p><b>2 mark</b></p> <ul style="list-style-type: none"><li>• correct solution.</li></ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• finds one inequality for <math>k</math></li></ul> |

|          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mks                                                                                  | Comments                                                                                                                                                                               |                                                                                       |
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| 14a(i)   | <p>Differentiating both sides:</p> $\frac{2x}{a^2} + \frac{2y}{b^2} \times \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ $m_T = -\frac{b}{a} \cot \theta$ $m_N = \frac{a}{b} \tan \theta$ <p>Equation of normal:</p> $y - b \sin \theta = \frac{a}{b} \tan \theta (x - a \cos \theta)$ $by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$ <p>Multiply both sides by <math>b \cos \theta</math></p> $by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$ $\sin \theta \cos \theta (a^2 - b^2) = ax \sin \theta - by \cos \theta$ $a^2 - b^2 = ax \sec \theta - by \operatorname{cosec} \theta$ | 2                                                                                    | <p><b>2 mark</b></p> <ul style="list-style-type: none"><li>• correct solution.</li></ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• finds <math>m_N</math></li></ul> |                                                                                       |
| 14a(ii)  | <p><math>G</math> occurs at the xint</p> $a^2 - b^2 = ax \sec \theta$ $x = \frac{a^2 - b^2}{a \sec \theta}$ $OG = \frac{a^2 - b^2}{a \sec \theta}$ $ON = a \cos \theta \text{ (x - ordinate of P)}$ $\frac{OG}{ON} = \frac{a^2 - b^2}{a \sec \theta} \times \frac{1}{a \cos \theta}$ $\text{but } b^2 = a^2(1 - e^2)$ $b^2 - a^2 = a^2 e^2$ $\frac{OG}{ON} = \frac{a^2 e^2}{a^2}$ $\frac{OG}{ON} = e^2$ $OG:ON = e^2:1$                                                                                                                                                                                                                                                                       | 2                                                                                    | <p><b>2 mark</b></p> <ul style="list-style-type: none"><li>• correct solution.</li></ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• finds <math>m_N</math></li></ul> |                                                                                       |
| 14a(iii) | <p>Since <math>\Delta PNG</math> and <math>\Delta PN'G'</math> are similar triangles and</p> $OG:ON = e^2$ $OG = e^2 \times ON$ $OG = e^2 \times PN'$ $GN = PN'(1 - e^2)$ $G'N' = PN \times (1 - e^2)$ $\therefore \text{Area } \Delta PN'G':\text{Area } \Delta PNG = 1:(1 - e^2)^2$                                                                                                                                                                                                                                                                                                                                                                                                         |  | 1                                                                                                                                                                                      | <p><b>1 mark</b></p> <ul style="list-style-type: none"><li>• correct ratio.</li></ul> |

|          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mks | Comments                                                                                                                                                                                                                                                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14b(i)   | $  \begin{aligned}  LHS &= \cos \frac{\theta}{2} \cos \frac{\theta}{4} \sin \frac{\theta}{4} \\  &= \cos \frac{\theta}{2} \left( 2 \times \cos \frac{\theta}{4} \sin \frac{\theta}{4} \right) \times \frac{1}{2} \\  &= \cos \frac{\theta}{2} \sin \frac{\theta}{2} \times 2 \times \frac{1}{2} \times \frac{1}{2} \\  &= \frac{1}{4} \sin \theta \\  &= RHS  \end{aligned}  $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1   | <b>1 mark</b><br>• correct proof.                                                                                                                                                                                                                                                                                                                 |
| 14b(ii)  | <p>Let <math>n = 1</math></p> <p>LHS = <math>\sin \theta</math></p> $  \begin{aligned}  RHS &= 2^1 \cos \frac{\theta}{2^1} \sin \frac{\theta}{2^1} \\  &= \sin \theta \quad (\text{double angle formula})  \end{aligned}  $ <p>Assume true for <math>n = k</math></p> $\sin \theta = 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \sin \frac{\theta}{2^k}$ <p>Prove true for <math>n = k + 1</math></p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;">       AIM: <math>\sin \theta = 2^{k+1} \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}}</math> </div> $  \begin{aligned}  RHS &= 2^{k+1} \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}} \\  &= 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \times 2 \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}} \\  &= 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \times \sin \frac{\theta}{2^k} \quad (\text{double angle formula}) \\  &= \sin \theta \quad (\text{if the assumption is true}) \\  &= LHS  \end{aligned}  $ <p>∴ If it is true for <math>n = k + 1</math>, if it is true for <math>n = k</math></p> <p>Since it is also true for <math>n = 1</math>, then it is true for positive integer of <math>n</math> by Mathematical Induction.</p> | 3   | <b>3 mark</b><br>• correct solution.<br><b>2 mark</b><br>• Completes all of the conditions below.<br><b>1 mark</b><br>• Completes one of the conditions below.<br><br><b>Conditions:</b><br><b>(A)</b> Tests for $n = 1$ , and writes the correct assumption<br><b>(B)</b> Shows the use of the assumption in a correct expression of $n = k + 1$ |
| 14b(iii) | $\lim_{n \rightarrow \infty} \left( \frac{\sin \left( \frac{\theta}{2^n} \right)}{\frac{\theta}{2^n}} \times 2^n \right) = \lim_{n \rightarrow \infty} \left( \frac{\sin \left( \frac{\theta}{2^n} \right)}{\left( \frac{\theta}{2^n} \right)} \right)$ <p>if <math>n \rightarrow \infty</math>, <math>\frac{\theta}{2^n} \rightarrow 0</math>, so <math>\frac{\theta}{2^n}</math> is a very small angle</p> $\lim_{n \rightarrow \infty} \left( \frac{\sin \left( \frac{\theta}{2^n} \right)}{\left( \frac{\theta}{2^n} \right)} \right) = \lim_{\left( \frac{\theta}{2^n} \right) \rightarrow 0} \left( \frac{\sin \left( \frac{\theta}{2^n} \right)}{\left( \frac{\theta}{2^n} \right)} \right) = 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1   | <b>1 mark</b><br>• Correct proof and <u>must explain</u> $\frac{\theta}{2^n}$ is a small angle.                                                                                                                                                                                                                                                   |

|         | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mks | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 14b(iv) | <p>Using the expression in part (ii), divide both side by <math>\theta</math></p> $\frac{\sin \theta}{\theta} = 2^n \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}}$ <p>As <math>n \rightarrow \infty</math>, ans using (iii)</p> $\lim_{n \rightarrow \infty} \left( \frac{\sin \theta}{\theta} \right) = \lim_{n \rightarrow \infty} \left( \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \times 2^n \right)$ $\frac{\sin \theta}{\theta} = \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \dots \times 1$ <p>By substituting <math>\theta = \frac{\pi}{2}</math></p> $\frac{\sin \frac{\pi}{2}}{\frac{\pi}{2}} = \cos \frac{\left(\frac{\pi}{2}\right)}{2} \cos \frac{\left(\frac{\pi}{2}\right)}{4} \cos \frac{\left(\frac{\pi}{2}\right)}{8} \dots$ $\frac{2}{\pi} = \cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \dots$                                                                                                                                                                                                                         | 3   | <p><b>3 mark</b></p> <ul style="list-style-type: none"> <li>• correct solution</li> </ul> <p><b>2 mark</b></p> <ul style="list-style-type: none"> <li>• Completes two of the conditions below.</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Completes one of the conditions below.</li> </ul> <p><b>Conditions:</b></p> <p><b>(A)</b> Uses <math>\theta = \frac{\pi}{2}</math></p> <p><b>(B)</b> Manipulates expression to use the limit in (iii)</p> <p><b>(C)</b> Uses (ii) and divides both sides by <math>\theta</math></p> |
| 14b(v)  | $\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ <p>we know: <math>\cos 2A = 2 \cos^2 A - 1</math></p> $\cos A = \sqrt{\frac{\cos 2A + 1}{2}}$ <div style="display: flex; justify-content: space-around;"> <div style="width: 45%;"> <math display="block">\cos \frac{\pi}{8} = \sqrt{\frac{\cos \frac{\pi}{4} + 1}{2}}</math> <math display="block">= \sqrt{\frac{\frac{\sqrt{2}}{2} + 1}{2}}</math> <math display="block">= \sqrt{\frac{\sqrt{2} + 2}{4}}</math> <math display="block">= \frac{\sqrt{\sqrt{2} + 2}}{2}</math> </div> <div style="width: 45%;"> <math display="block">\cos \frac{\pi}{16} = \sqrt{\frac{\cos \frac{\pi}{8} + 1}{2}}</math> <math display="block">= \sqrt{\frac{\frac{\sqrt{\sqrt{2} + 2}}{2} + 1}{2}}</math> <math display="block">= \sqrt{\frac{\sqrt{\sqrt{2} + 2} + 2}{4}}</math> <math display="block">= \frac{\sqrt{\sqrt{\sqrt{2} + 2} + 2}}{2}</math> </div> </div> $\therefore \frac{\pi}{2} = \frac{2}{\sqrt{2}} \times \frac{2}{\frac{\sqrt{\sqrt{2} + 2}}{2}} \times \frac{2}{\sqrt{\sqrt{\sqrt{2} + 2} + 2}} \dots \dots$ $\therefore \pi = 2 \left( \frac{2}{\sqrt{2}} \times \frac{2}{\frac{\sqrt{\sqrt{2} + 2}}{2}} \times \frac{2}{\sqrt{\sqrt{\sqrt{2} + 2} + 2}} \dots \dots \right)$ | 2   | <p><b>2 mark</b></p> <ul style="list-style-type: none"> <li>• correct solution, showing upto <math>\cos \frac{\pi}{8}</math>.</li> </ul> <p><b>1 mark</b></p> <ul style="list-style-type: none"> <li>• Finds the exact value of <math>\cos \frac{\pi}{4}</math> and attempt to use the double angle to find <math>\cos \frac{\pi}{8}</math></li> </ul>                                                                                                                                                                                                      |